

a- Simplify the following expression using K-Map: .

- 1- $F(x, y, z) = \sum(2, 3, 6, 7)$
- 2- $F(x, y, z, w) = \sum(1, 4, 5, 6, 7, 13)$
- 3- $F(x, y, z, w) = \sum(1, 3, 8, 10, 15)$, $d(x, y, z, w) = \sum(0, 2, 9)$
- 4- $F(x, y, z, w) = \sum(4, 5, 7, 12, 13, 14)$, $d(x, y, z, w) = \sum(1, 9, 11, 15)$

b- Simplify the following expression using K-Map:

1- $X'Y' + YZ + X'YZ'$

2- $A'B'C'D' + AC'D' + B'CD' + A'BCD + BC'D$

c- Implement this expression using NAND gate only and another time using NOR gate only:

$$F = XY + X'Y' + Y'Z$$

d- Convert the following logic gate circuit into a Boolean expression and construct the truth table.

